Polar Coordinates and Conversions

Converting points

1 Convert each polar point (r, θ) to rectangular coordinates (exact values):

a $\left(4, \frac{\pi}{3}\right)$ **b** $\left(2, \frac{3\pi}{4}\right)$

2 Convert each rectangular point to polar coordinates with $r \geq 0$ and $0 \leq \theta < 2\pi$:

a $(-3, 3)$ **b** $(0, -5)$

Converting equations

3 Convert the polar equation $r = 6\cos\theta$ to rectangular form and identify the curve.

4 Convert each rectangular equation to polar form:

a $x^2 + y^2 = 16$ **b** $y = x$

5 Convert the polar equation $r = \frac{2}{1 - \sin\theta}$ to rectangular form.

Multiple representations

6 Give two other pairs (r, θ) that represent the same point as $\left(3, \frac{\pi}{4}\right)$: one with $r > 0$ and θ outside $[0, 2\pi)$, and one with $r < 0$.

7 Plot-reasoning: explain why the polar point $\left(-2, \frac{\pi}{2}\right)$ corresponds to the same location as $\left(2, \frac{3\pi}{2}\right)$.

Distance between polar points

8 Convert $\left(4, \frac{\pi}{6}\right)$ and $\left(4, \frac{2\pi}{3}\right)$ to rectangular coordinates, then find the distance between them, to two decimal places.

Further conversion practice

9 Convert each polar point to rectangular coordinates (exact values):

a $(6, \pi)$ **b** $\left(5, \frac{5\pi}{6}\right)$

10 Convert the polar equation $r^2 = 4\sin 2\theta$ to note its symmetry, and identify it as a standard curve type.